3/10: 教材189-190页5,6; 丘老师上册181页3; 下册126页2,3.
3/13: 197-198页1-8,10,11.
f_A denotes the characteristic polynomial of A, and p_A denotes its minimal polynomial.

189-5

Let

$$A = \begin{pmatrix} 6 & -3 & -2 \\ 4 & -1 & -2 \\ 10 & -5 & -3 \end{pmatrix}$$

Is A similar over the field $\mathbb{R}$ to a diagonal matrix? Is it over $\mathbb{C}$?
Solution:

$$f_A(x) = \det(xI - A) = \det \begin{pmatrix} x-6 & 3 & 2 \\ -4 & x+1 & 2 \\ -10 & 5 & x+3 \end{pmatrix} = (x^2+1)(x-2).$$

Since f_A does not split over $\mathbb{R}$, it is not diagonalizable. $f_A = (x+i)(x-i)(x-2)$ has distinct roots in $\mathbb{C}$, so it is diagonalizable.

189-6

Let $T \in \mathcal{L}(\mathbb{R}^4)$ be such that

$$T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{pmatrix}$$

Under what conditions on a, b, c is T diagonalizable.
Solution: $f_T(x) = x^4$, so T is diagonalizable iff $\mathrm{dimKer}T = 4$, so T is diagonalizable only if $a = b = c = 0$.

丘181-3

Calculate A^m:
(1) $A = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$; (2) $\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$.
Solution: (1) $f_A(x) = (x-2)(x-3)$, and $(2,1)^T \in \mathrm{Ker}(A-2I)$, $(1,1)^T \in \mathrm{Ker}(A-3I)$. Let $P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, then $A = P^{-1}\mathrm{diag}(2,3)P$, so $A^m = P^{-1}\mathrm{diag}(2^m, 3^m)P$, and

$$A^m = \begin{pmatrix} 2^{m+1} - 3^m & -2^{m+1} + 2 \cdot 3^m \\ -2^m + 3^m & -2^m + 2 \cdot 3^m \end{pmatrix}.$$

(2) $f_A(x) = (x-2)(x+1)$, and $(1,1)^T \in \mathrm{Ker}(A-2I)$, $(2,-1)^T \in \mathrm{Ker}(A+I)$. Hence

$$A^m = P^{-1}\mathrm{diag}(2^m, (-1)^m)P = \frac{1}{3}\begin{pmatrix} 2^m + 2(-1)^m & 2^{m+1} - 2(-1)^m \\ 2^m - (-1)^m & 2^{m+1} + (-1)^m \end{pmatrix}.$$

丘126-2

Is A diagonalizable?
(1) $A = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}$; (2) $A = \begin{pmatrix} 2 & 3 & 2 \\ 1 & 8 & 2 \\ -2 & -14 & -3 \end{pmatrix}$.
Solution: (1) $f_A(x) = (x-1)^2(x-10)$, and $\mathrm{rank}(A - I) = 1$ so $\mathrm{dimKer}(A - I) = 2$. Hence A is diagonalizable.
(2) $f_A(x) = (x-3)^2(x-1)$, and $\mathrm{rank}(A - 3I) = 2$ so $\mathrm{dimKer}(A - 3I) = 1$ and A is not diagonalizable.

丘126-3

Suppose

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

(1) Find all eigenvalues and eigenvectors of A;
(2) Find a basis of V such that A is diagonal under that basis.
Solution: (1) $f_A(x) = x^2(x-1)^2$ so the eigenvalues are $0, 1$ and the eigenvectors are $(0, y, z, 0)$ and $(x, 0, x, w)$.
(2) Consider $e_1 = (0,1,0,0), e_2 = (0,0,1,0), e_3 = (1,0,1,0), e_4 = (0,0,0,1)$.

197-1

Let V be a finite dimensional vector space. What is the minimal polynomial of 1_V? What is the minimal polynomial of 0_V?
Solution: The minimal polynomial for 1_V is $x - 1$, and that of 0_V is x.

198-2

Let $a, b, c \in F$, and

$$A = \begin{pmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{pmatrix}.$$

Prove that the characteristic polynomial for A is $x^3 - ax^2 - bx - c$ and that this is also the minimal polynomial for A.

Proof: $f_A(x) = \det \begin{pmatrix} x & 0 & -c \\ -1 & x & -b \\ 0 & -1 & x-a \end{pmatrix} = x^3 - ax^2 - bx - c$. Note that

$$A^2 = \begin{pmatrix} 0 & c & ac \\ 0 & b & c+ab \\ 1 & a & b+a^2 \end{pmatrix}$$

so if $uA^2 + vA + wI = 0$, then considering the first column we infer $u = v = w = 0$. Hence $\deg p_A \geqslant 3$, so $p_A = f_A = x^3 - ax^2 - bx - c$.

198-3

Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -2 & -2 & 2 & 1 \\ 1 & 1 & -1 & 0 \end{pmatrix}$$

Show that the characteristic polynomial for A is $x^2(x-1)^2$ and that it is also the minimal polynomial.

Proof: $f_A(x) = \det(xI - A) = x^2(x-1)^2$. Note that the characteristic polynomial for $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ is x^2, hence the minimal polynomial is also x^2. Likewise the minimal polynomial for $\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ is $(x-1)^2$, so the minimal polynomial p_A satisfy $x^2(x-1)^2 \mid p_A$. Therefore $p_A = f_A = x^2(x-1)^2$.

198-4

Is the matrix of Exercise3 similar over $\mathbb{C}$ to a diagonal matrix?

Proof: No, since 0 is not a simple root of p_A.

198-5

Let V be an $n-$dimensional vector space and let $T \in \mathcal{L}(V)$. Suppose there exists some positive integer k such that $T^k = 0$. Prove that $T^n = 0$.

Proof: Suppose the minimal polynomial of T is p_T, then by Hamilton-Cayley theorem $\deg p_T \leqslant n$. Since $T^k = 0, p_T \mid x^k$ so $p_T = x^l$ for some $l \leqslant n$. Hence $p_T \mid x^n$ so $T^n = 0$.

198-6

Find a 3×3 matrix for which the minimal polynomial is x^2.

Solution: Consider $A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, then $A^2 = 0$ but $uA + vI \neq 0$ for all u, v. Hence the minimal polynomial for A is x^2.

198-7

Let n be a positive integer, and V be the space of polynomials over $\mathbb{R}$ with degree at most n. Let D be the differentiation operator on V. What is the minimal polynomial for D?

Solution: V has a basis $\mathcal{B} = \{x^n, \cdots, x, 1\}$, and

$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & \cdots & & \\ n & 0 & \cdots & & \\ 0 & n-1 & \cdots & & \\ & & \cdots & & \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

Since $f_D = x^{n+1}$ and $D^n(x^n) = n! \neq 0$, the minimal polynomial is $p_D = x^{n+1}$.

198-8

Let P be the operator on $\mathbb{R}^2$ which projects each vector onto the $x-$axis, parallel to the $y-$axis: $P(x, y) = (x, 0)$. Show that P is linear. What is the minimal polynomial for P?

Proof: Clearly P is linear. Note that $x^2 - x$ annihilates P, and $P, P - I \neq 0$, so the minimal polynomial is $x^2 - x$.

198-10

Let $V = F^{n \times n}$. Let A be a fixed $n \times n$ matrix, and $T \in \mathcal{L}(V)$ be $T(B) = AB$. Show that the minimal polynomial for T is the minimal polynomial for A.

Proof: Note that $p(T)(B) = p(A) \cdot B$ for any polynomial p, hence $p_A(T) = 0$ and $p_T(A) = 0$, so $p_A = p_T$.

198-11

Let $A, B \in F^{n \times n}$. Recall that AB, BA have the same eigenvalues. Do they have the same characteristic polynomial? Do they have the same minimal polynomial?

Solution: Yes for characteristic polynomials, no for minimal polynomials.

Consider $A : (x_1, \cdots, x_n) \mapsto (x_1, 0, 0, \cdots, 0)$ and $B : (x_1, \cdots, x_n) \mapsto (0, x_1, 0, \cdots, 0)$, then $AB = 0$ while $BA = B \neq 0$. Hence the minimal polynomial of AB is x, while that of BA isn't (its actually x^2).

However, note that

$$\det(xI - BA) = \det \begin{pmatrix} xI - BA & 0 \\ B & I \end{pmatrix} = \det \begin{pmatrix} xI & A \\ B & I \end{pmatrix} = \det \begin{pmatrix} xI - AB & 0 \\ B & I \end{pmatrix} = \det(xI - AB).$$

Hence they have the same characteristic polynomial.